

Portfolio Management

Guideline: The uploaded dataset (.csv) contains monthly returns of U.S. stocks from January 2000 to December 2020. Using this dataset and any software of your choice (Python is strongly recommended), your team must design and backtest a monthly rebalanced portfolio. The submission must include: (1) Project Report(.pdf) and (2) Programming codes.

(0) Submission to LMS

- Submit a single compressed file in the following format (non-compliance = -2 points)
 - *Team#_PA1.zip*

(1) Project Report

- Recommended table of contents:
 - 1) Overview
 - Summarize your team's work and key findings.
 - MUST include a process diagram illustrating the overall design of your portfolio investment strategy.
 - 2) Portfolio management framework
 - Provide a detailed explanation of your portfolio investment strategy, including asset/security selection, portfolio optimization models, and the use of time-series models (if applicable).
 - 3) Performance Evaluation
 - Include figure(s) showing the portfolio value during the backtesting period. – Provide table(s) reporting portfolio performance with annualized return, annualized standard deviation, Sharpe ratio, and maximum drawdown (additional metrics are welcome). – Compare your portfolio against the S&P 500 index as the baseline benchmark.
- Convert the report to PDF format before submission.

(2) Programming Code

- Submit the original source files (.py or .ipynb). R users may submit R packages.
- Ensure that every line of code is documented with comments (in either Korean or English).

[Data Description]

The dataset comes from the **Center for Research in Security Prices (CRSP)** and contains **monthly U.S. stock data** from January 2000 to December 2020. Each row represents one stock in a given month with the following variables:

- **PERMNO:** Unique permanent identifier for each stock. Always use PERMNO as the stock ID.
- **date:** Last trading day of the month.
- **EXCHCD:** Exchange code: 1 = NYSE, 2 = AMEX, 3 = NASDAQ. Use only these exchanges.
- **TICKER** and **COMNAM:** Ticker and company name (for reference only, not unique).
- **ADJ_PRC:** Adjusted closing price, accounting for splits and other corporate actions.
- **VOL:** Monthly trading volume (not adjusted for splits). Do not multiply by price for dollar volume.

Key Notes

- Use **PERMNO** as the stock identifier.
- Focus on **NYSE, AMEX, and NASDAQ** stocks.
- Use **ADJ_PRC** for return calculations.
- Treat **VOL** as an activity measure, not a value measure.

[Portfolio Strategy]

Drawing on existing research (or your own innovative ideas if they offer clear merit), your team is expected to design and implement a portfolio strategy. At minimum, the portfolio should satisfy at least one of the following requirements:

- **Security Selection and Mean-Variance Portfolio (Minimum Requirement):** Select a set of securities and construct a mean-variance optimized portfolio (GMVP, MSRP, and etc).
- **Time-Series Modeling:** You may incorporate time-series models to forecast returns and/or volatilities. If you team is focusing on volatility forecasting modeling, you must show the corresponding volatility-managed portfolio performance.
- **Factor Model-Based Investment:** An exclusive portfolio based on factor model is also acceptable; an equal-weight allocation is sufficient.